

Ujaan Das

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EDUCATION

University of Warwick

Master of Science in Computer Science

Hong Kong University of Science and Technology

Bachelor of Engineering in Computer Engineering

Coventry, United Kingdom

Sep. 2025 – Sep. 2026

Hong Kong SAR

Sep. 2021 – May 2025

EXPERIENCE

Software Engineering Intern (Part Time)

Jan. 2025 – Aug. 2025

Stellerus Technology

Hong Kong SAR

- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%
- Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions

Software Engineering Intern

May 2023 – Aug. 2023

Stellerus Technology

Hong Kong SAR

- Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

Research Assistant

Jun. 2022 – Aug. 2022

HCI Initiative, HKUST

Hong Kong SAR

- Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments
- Collaborated with post-docs to build a modular React + TypeScript frontend using custom hooks and Tailwind CSS, improving component reusability and streamlining development across multiple researchers
- Contributed to FARPLS, a feature-augmented preference learning system that improved label quality and user engagement without increasing cognitive load (published at ACM/IUI'24)

PROJECTS

MSc Dissertation - Invariants | C++, Gtest, CMake, Python, Pybind11, Nix

[Github](#)

- Engineering a specification DSL using semantic invariants and refinement types to guide LLM decoding via verifiable runtime constraints
- Architecting a C++ compilation runtime with Python bindings for seamless integration into inference pipelines, enforcing schema compliance with targeted <30ms overhead

Snip | C++, Termios, Gtest, CMake, Nix

[Github](#)

- Architected an async C++ TUI framework and Vim-style modal editor with a thread-safe, non-blocking event pipeline and background worker pools for deferred execution
- Integrated native LSP tooling for real-time diagnostics and built hermetic, reproducible environments via Nix for cross-platform consistency

oojsite | Go, HTML, TailwindCSS, Nix

[Github](#)

- Built a minimalist static site generator in Go (text/template) that compiles Markdown into composable HTML with reusable components, enabling fast, dependency-free blog generation

TerraViz | Next.js, React, TypeScript, CesiumJS, TailwindCSS

[DevPost](#)

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data
- Developed a 3D CesiumJS visualization layer to render multi-source geospatial data into actionable maps

SKILLS

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: GTest, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic, Celery

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git